V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning

Mur-Labadia, Lorenzo, Muckley, Matthew, Bar, Amir, Assran, Mido, Sinha, Koustuv, Rabbat, Mike et al.

https://arxiv.org/abs/2603.14482

Research Questions

- How can self-supervised video models learn dense spatial features while retaining global scene understanding?
- Why do masked prediction objectives in V-JEPA produce poor dense representations and how can this be fixed?
- Can unified training across images and videos with multi-modal tokenizers improve dense feature quality?

Core Idea

The core problem is that existing V-JEPA models learn good global representations but poor dense spatial features because the prediction loss only applies to masked tokens, allowing visible context tokens to aggregate global information rather than encode local structure. The key insight is to extend the predictive loss to ALL tokens (both masked and visible) using a weighted context loss, forcing each token to encode local spatial-temporal information. This dense predictive loss combined with deep self-supervision at multiple encoder layers produces representations that are both spatially structured and semantically coherent.

Key Formula

Context Loss

$$L_{context} = \frac{1}{|C|} \sum_{i \in C} \lambda_i \| P_\phi(E_\theta(x), \Delta_y)_i - sg(E_\theta(y)_i) \|_1$$

C is the set of context (visible) token indices, λ_i is a patch-specific weighting parameter, P_φ is the predictor, E_θ is the encoder, sg is stop-gradient. This loss supervises visible context tokens to prevent them from becoming global aggregators and forces local spatial encoding.

Dynamic Weighting

$$\lambda_i = \frac{\lambda}{\sqrt{d_{min}(i, M)}}$$

λ is a hyperparameter, d_min(i,M) is the minimum distance between context token i and closest masked token in M. This weights context tokens inversely by distance to masked regions, emphasizing local continuity between masked and unmasked areas.

Key Figure

arXiv:2603.14482v2 [cs.CV] 17 Mar 2026

V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning

Lorenzo Mur-Labadia[1,2,*], **Matthew Muckley**[1], **Amir Bar**[1], **Mido Assran**[1], **Koustuv Sinha**[1], **Mike Rabbat**[1], **Yann LeCun**[1], **Nicolas Ballas**[1,†], **Adrien Bardes**[1,†]

[1]FAIR at Meta, [2]Universidad de Zaragoza
[*]Work done at Meta, [†]Joint last author

We present **V-JEPA 2.1**, a family of self-supervised models that learns *dense*, high-quality representations for visual scenes in both images and videos, while retaining strong *global* scene understanding. V-JEPA 2.1 combines four key ingredients: (i) a **Dense Predictive Loss**, a masking-based objective in which *all* tokens—visible context and masked tokens alike—contribute to the training loss, encouraging explicit spatial and temporal grounding; (ii) **Deep Self-Supervision**, which applies the self-supervised objective hierarchically at multiple intermediate encoder layers to improve representation quality ; (iii) **Multi-Modal Tokenizers** that support unified training over images and videos; and (iv) effective **model and data scaling**. These design choices substantially improve dense feature quality, yielding representations that are spatially structured, semantically coherent, and temporally consistent. . Empirically, V-JEPA 2.1 achieves state-of-the-art results on a range of benchmarks: 7.71 mAP on Ego4D for short-term object-interaction anticipation, 40.8 Recall@5 on EPIC-KITCHENS for high-level action anticipation, and a 20% improvement in real-robot grasping success rate over VJEPA-2 AC. The model also demonstrates state-of-art performances in robotic navigation (5.687 ATE on Tartan Drive), depth estimation (0.307 RMSE on NYUv2 with a linear probe), and global recognition (77.7% on Something-Something-V2). Our results demonstrate that V-JEPA 2.1 advances the state of the art in dense visual understanding and world modeling.

Date: March 18, 2026
Correspondence: lmur@unizar.es, abardes@meta.com
Code: https://github.com/facebookresearch/vjepa2

Meta

Figure 1 V-JEPA 2.1 unlocks high-quality dense features. We compute PCA on patch features extracted from the same image or video and map the top three components to RGB channels for both V-JEPA 2 (ViT-g) and V-JEPA 2.1 (ViT-G). Our novel V-JEPA 2.1 produces dense representations with strong spatial and temporal consistency, learning semantically coherent features where similar objects map to the same PCA components.

Figure 1: Shows PCA visualizations comparing V-JEPA 2 vs V-JEPA 2.1 dense features, demonstrating how V-JEPA 2.1 produces coherent spatial structure where similar objects map to same components.

Limitations

- Computational overhead from deep self-supervision at multiple encoder layers
- Dynamic weighting scheme adds hyperparameter sensitivity for different datasets
- Multi-modal tokenizer requires careful balancing between image and video data distributions
- Performance on some global tasks (video classification) shows only modest improvements over V-JEPA 2